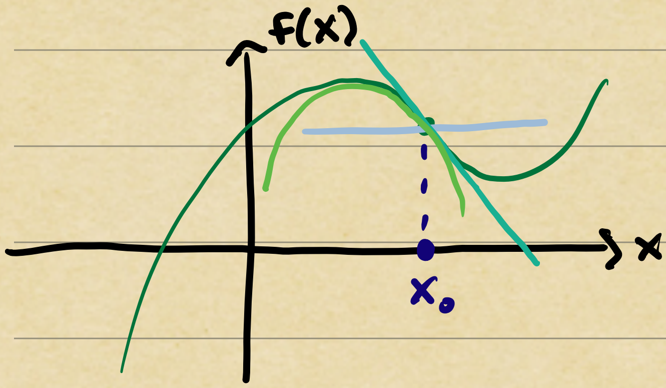
Intro to Linear Approximation

Idea: build complex functions from simple ones.

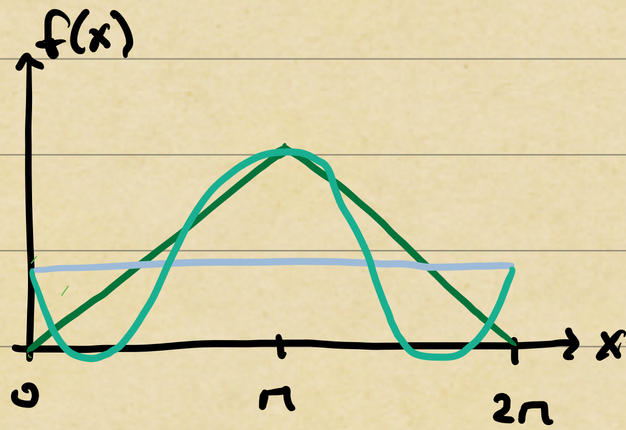
Example 1: Taylor Polynomials



$$f(x) \approx f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n$$

Build up $f(x)$ near x_0 using monomials $(x-x_0)^j$.

Example 2: Fourier Series



$$f(x) \approx \frac{a_0}{2} + a_1 \cos(x) + a_2 \cos(2x) + \cdots + a_n \cos(nx)$$

discuss coeffs later

Build up $f(x)$ using trigonometric functions.

Q: How accurate are these approximations?

Remainder Formula (Taylor Series)

Taylor Polynomial $f_n(x) = f(x_0) + f'(x_0)(x-x_0) + \cdots + \frac{1}{n!} f^{(n)}(x_0)(x-x_0)^n$

Claim: If f has $n+1$ continuous derivatives in a neighborhood $I_\delta = [x_0-\delta, x_0+\delta]$ of x_0, then $(n \geq 1)$

$$(R) \quad f(x) - f_n(x) = \frac{1}{n!}\int_{x_0}^{x} f^{(n+1)}(t)(x-t)^n\,dt, \quad x \in I_\delta.$$

The remainder formula allows us to estimate

max error

local improvement

$$(E) \quad \sup_{x\in I_\delta} |f(x) - f_n(x)| \leq \frac{\delta^{n+1}}{(n+1)!} \sup_{x \in I_\delta} |f^{(n+1)}(x)|.$$

"size" of local fluctuations

pf| The proof combines the fundamental theorem of calculus with integration-by-parts.

Key Idea 1: Fundamental thm. of calc. (F.T.C.)

$$f(x) = f(x_0) + \int_{x_0}^{x} f'(t)\,dt$$

Key Idea 2: Integration-by-parts. (I.B.P.)

$$f(x) = f(x_0) + \left[t f'(t)\right]_{x_0}^{x} - \int_{x_0}^{x} f''(t)\, t\, dt$$

$$= f(x_0) + x f'(x) - x_0 f'(x_0) - \int_{x_0}^{x} f''(t)\, t\, dt$$

Now, we apply F.T.C. again to simplify.

$$x f'(x) = x f'(x_0) + x \int_{x_0}^{x} f''(t)\, dt$$

$$= f(x_0) + f'(x_0)(x - x_0) + \int_{x_0}^{x} f''(t)(x-t)\, dt$$

Now, repeatedly apply I.B.P. to integral remainder. E.g., for the $n=2$ case we proceed to calculate

$$\int_{x_0}^{x} f''(t)(x-t)\, dt = \frac{1}{2} f''(x_0)(x-x_0)^2 + \frac{1}{2}\int_{x_0}^{x} f^{(3)}(t)(x-t)^2\, dt$$

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2} f''(x_0)(x-x_0)^2 + \frac{1}{2}\int_{x_0}^{x} f^{(3)}(t)(x-t)^3\, dt$$

To prove the general case, proceed by induction. Suppose that (R) holds for $n = k \geq 1$, we'll show that it must also hold for $k+1$.

Since we already know it holds for $n=1$,
this implies it also holds for $n=2,3,4,\dots$

$$f(x_0) = f_k(x_0) + \frac{1}{k!}\int_{x_0}^{x} f^{(k+1)}(t)(x-t)^k\,dt$$

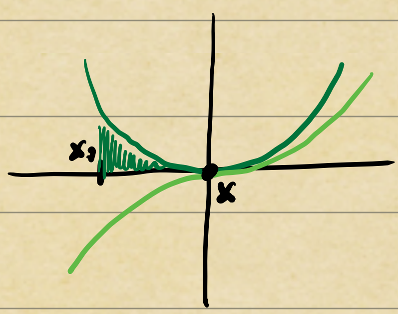
$$\int_{x_0}^{x} f^{(k+1)}(t)(x-t)^k\,dt = \frac{1}{k+1} f^{(k+1)}(x_0)(x-x_0)^{k+1} + \frac{1}{k+1}\int_{x_0}^{x} f^{(k+2)}(t)(x-t)^{k+1}\,dt$$

$$\Rightarrow \quad f(x) = f_{k+1}(x) + \frac{1}{(k+1)!}\int_{x_0}^{x} f^{(k+2)}(t)(x-t)^{k+1}\,dt$$

This establishes the remainder formula.
Now, let's estimate the size of the remainder.

$$\sup_{x\in I_\delta} |f(x) - f_n(x)| \leq \frac{1}{n!}\sup_{x\in I_\delta}\left|\int_{x_0}^{x} f^{(n+1)}(t)(x-t)^n\,dt\right|$$

$$\leq \frac{1}{n!}\sup_{t\in I_\delta}|f^{(n+1)}(t)|\int_{x_0}^{x}|x-t|^n\,dt$$



$$\leq \frac{1}{n!}\sup_{x\in I_\delta}|f^{(n+1)}(x)|\,\frac{|x-x_0|^{n+1}}{n+1}$$

$$\leq \frac{\delta^{n+1}}{(n+1)!}\sup_{x\in I_\delta}|f^{(n+1)}(x)|.$$

HW!

Q: What is the analogous estimate for Fourier?

Key Observations about Taylor Series:

1) The Taylor polynomial always interpolates f at x_0, e.g., $f(x_0) = f_n(x_0)$.

2) The order of interpolation is n, i.e., from (R)-(E), we have that

$$f(x) - f_n(x) \leq \underbrace{C}_{\text{indep. of } x} |x - x_0|^{n+1}, \quad \text{for } x \text{ near } x_0.$$

We write $|f(x) - f_n(x)| = O(|x-x_0|^n)$ as $x \to x_0$.

3) If all derivatives of $f(x)$ on I_δ exist and grow slower than $(n+1)!/\delta^{n+1}$, then f converges uniformly to f on I_δ, i.e.,

$$\sup_{x \in I_\delta} |f(x) - f_n(x)| \to 0 \quad \text{as} \quad n \to \infty.$$

Note: Taylor series may not converge (to f!) as $n \to \infty$ for all $x \in I_\delta$, even when $f \in C^\infty(I_\delta)$.

Example: Try $f(x) = \exp(-1/x^2)$ at $x_0 = 0$!

Convergence of Fourier Series

$$f_N(x) = \frac{1}{\sqrt{2}} \sum_{k=-N}^{+N} \hat{f}_k e^{i\pi k x}, \qquad \hat{f}_k = \frac{1}{\sqrt{2}} \int_{-1}^{+1} f(x) e^{-i\pi k x}\, dx.$$

Thm If $f, f', \ldots, f^{(n)}$ are continuous and periodic on $[-1,1]$ with $\sup_{x\in[-1,1]} |f^{(n)}(x)| \le M$, then

$$(n \ge 1) \qquad |\hat{f}_k| \le \frac{\sqrt{2}\, M}{(\pi k)^n} \qquad k = \pm 1, \pm 2, \ldots$$

The truncation error for f_N satisfies

$$(n \ge 2) \qquad |f(x) - f_n(x)| \le \frac{2\sqrt{2}\, M}{(n-1)\pi^n N^{n-1}}$$

Pf See Homework 1, Problem 2

(Hint: Integrate-by-Parts)

Note: For certain functions with piecewise continuous derivatives, like $|x|$, the theorem can be sharpened to provide $\mathcal{O}(k^{-n-1})$ coeff. decay and $\mathcal{O}(N^{-n})$ trunc. err.

Key Observations about Fourier Series:

1) Unlike Taylor series, Fourier Series converge uniformly for all continuously differentiable, periodic functions.

2) The rate of convergence for Fourier Series (as $N \to \infty$) improves with each additional continuous derivative of f.

Theme: "Smooth" functions, with many continuous derivatives of modest growth, are often well-approximated by linear combinations of simple functions like polynomials. In contrast, "rough" functions, with singularities or fewer continuous derivatives, may suffer from slow convergence rates or non-convergence.

Q: How can we systematically reason about approximation/truncation errors?